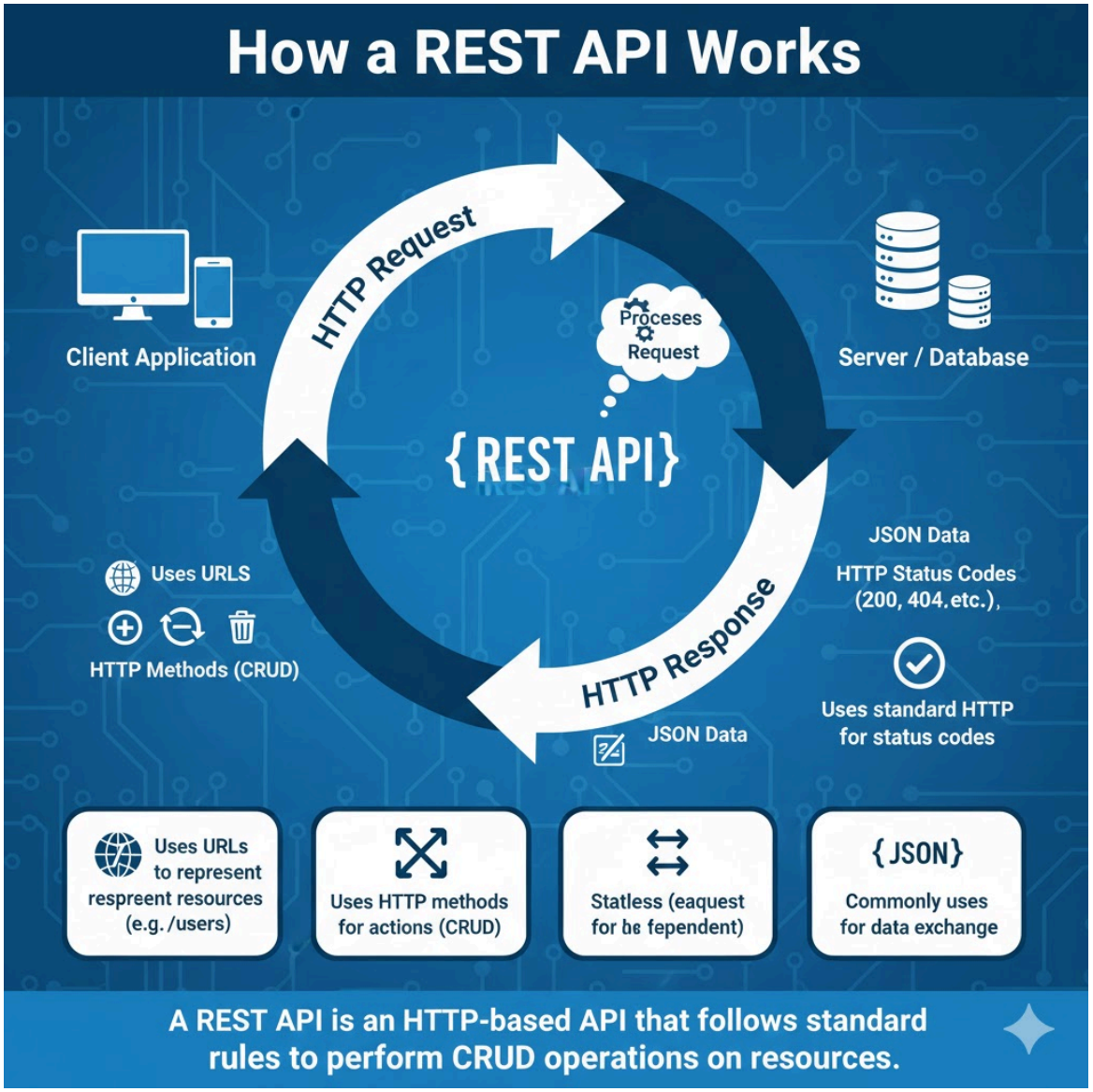
22. REST API and AXIOS Library

▼ What is a REST API?

- A **REST API** stands for `Representational State Transfer API`.
- A REST API is a way to communicate client to the server over the internet using **HTTP methods** like **GET, POST, PUT/PATCH, and DELETE**.
- It follows simple rules:
  - Uses **URLs to represent resources** (e.g., `/users`, `/products/1`)
  - Uses **HTTP methods for actions** (CRUD)
  - Is **stateless** (each request is independent)
  - Commonly uses **JSON** for data exchange
  - Uses **standard HTTP status codes**

👉 **In one line:**

> **A REST API is an HTTP-based API that follows standard rules to perform CRUD operations on resources.**



▼ What is `Axios`?

- **Axios** is a **JavaScript HTTP client library.**
- Axios library is used to send HTTP requests from a React application to a server or REST API.
- It is a Promise-based API.
- It works in **browser and Node.js**
- It is mainly used to **consume REST APIs**
- It supports all HTTP methods like **GET, POST, PUT, PATCH, DELETE**

▼ Difference Between Axios and fetch

Why Axios is Better than Fetch?

Feature	Axios	Fetch
JSON handling	Automatic	Manual (`res.json()`)
Error handling	Built-in	Must handle manually
Syntax	Clean & short	More boilerplate
Request timeout	Supported	Not supported
Interceptors	Yes	No
Default headers	Easy	Manual
Browser support	Excellent	Native

👉 That's why Axios is preferred in React applications.

▼ Command to Install Axios

💡 npm install axios

▼ Axios with `async / await`

▼ Base URL Example

```
const BASE_URL = "https://api.example.com/users";
```

▼ 1. `GET` – Read Data

- Get All Records.
- **Syntax**:

```
const getUsers = async () => {
  try {
    const response = await axios.get(BASE_URL);
    console.log(response.data);
  } catch (error) {
    console.log(error);
```

```
  }
};
```

- **Get Single Record: URL with ID**

```
const getUserById = async (id) => {
  try {
    const response = await axios.get(`${BASE_URL}/${id}`);
    console.log(response.data);
  } catch (error) {
    console.log(error);
  }
};
```

▼ 2. `POST` – Create Data

- **Syntax**:

```
const addUser = async () => {
  try {
    const response = await axios.post(BASE_URL, {
      name: "John",
      age: 20
    });
    console.log(response.data);
  } catch (error) {
    console.log(error);
  }
};
```

- `Note`: *POST creates a **new resource**, so the server generates the ID.*

▼ 3. `PUT` – Replace Data

- PUT replaces the **specific resource identified by ID**.
- **Syntax**:

```
const updateUserPut = async (id) => {
  try {
    await axios.put(`${BASE_URL}/${id}`, {
      name: "John",
      age: 20
    });
  } catch (error) {
    console.log(error);
  }
};
```

▼ 4. `PATCH` – Update Partial Data (Recommended)

- PATCH updates **only selected fields** of a specific record.
- **Syntax**:

```
const updateUserPatch = async (id) => {
  try {
    await axios.patch(`${BASE_URL}/${id}`, {
      age: 20
    });
  } catch (error) {
    console.log(error);
  }
};
```

▼ 5. `DELETE` – Remove Data

- DELETE must know **which resource to remove**.
- **Syntax**:

```
const deleteUser = async (id) => {
  try {
    await axios.delete(`${BASE_URL}/${id}`);
  } catch (error) {
    console.log(error);
```

```
  }
};
```

▼ Final Overview:

> ***In REST APIs, GET and POST usually use the same URL, but UPDATE and DELETE always require an ID in the URL.***